VIRTUAL REALITY SIMULATION (VR)

OVERVIEW

Virtual Reality (VR) is the representation of complex scientific and/or technical concepts in a visual form. Applying leadership and 21st century skills, participants use video and 3D computer graphics tools and design processes to communicate, inform, analyze, and/or illustrate a given topic, idea, subject, or concept based on the theme posted on the TSA website under *Themes & Problems*.

ELIGIBILITY

One (1) team per chapter may participate; individual entries are permitted.

TIME LIMITS

PRELIMINARY ROUND

A. Recording of the simulation must be two to three (2-3) minutes in length.

B. A five (5) point deduction will be applied for every fifteen (15) second interval that the simulation is shorter than two (2) minutes or longer than three (3) minutes.

C. Video of the simulation time length is calculated from the start of the first image or sound to the end of the last image or sound.

D. All components of the chapter's entry (Simulation URL and documentation PDF) must be finished, submitted, and accessible via the Internet by 11:59pm ET on a designated date mid-May.

SEMIFINAL ROUND

A. Up to four (4) minutes to setup/load of the presentation

B. Up to six (6) minutes to view the presentation – Three (3) minutes per judge.

C. Four (4) minutes are allowed for the interview.

ATTIRE

TSA competition attire is required for this event.

PROCEDURE

PRE-CONFERENCE

A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules.

B. Participants review the Use of Generative Artificial Intelligence (GenAI) found in the General Rules.

C. Participants concentrate their efforts on creating video and 3D computer graphic that illustrates a given topic.

D. Simulation URL, and documentation portfolio PDF must be submitted by 11:59pm ET on a designated date in mid-May.

PRELIMINARY ROUND

A. Entries are reviewed by judges with neither students nor advisors present.

1. Judges review and score the Video of the Simulation criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted.
2. Judges score the Documentation Portfolio criteria of those top twenty-four (24) contestants to determine the top twelve (12) semifinalist teams.

B. A list of twelve (12) semifinalists (in random order) is posted.

SEMIFINAL ROUND

A. No more than two (2) semifinalist team representatives report at the time and place stated in the conference program to sign up for a presentation/interview time.

B. No more than two (2) representatives from each semifinalist team report at the assigned time and place for the presentation/interview.

C. Each semifinalist team are to present their VR presentation using a VR technology of their choice. Semifinalists are responsible to provide the hardware for the visualization.

D. Semifinalist will answers questions about their portfolio from the judges, discussing the purpose, value, research, and design process.

E. The top ten (10) finalists are announced at the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

A. All entries must be the original work of the participant or team.
   1. Where applicable, all ideas, text, images, and sound from other sources must be cited.
   2. If copyrighted material is used, proper written permission must be included.
   3. Failure to follow this procedure results in disqualification.

B. All entries become the property of TSA and will not be returned after judging.

C. Documentation Portfolio:
   1. Portfolio must include the following pages in a multi-page PDF document in this order:
      a. Title page with the event title, the conference city and state, and the year; one (1) page
      b. Table of contents; pages as needed
      c. Purpose of simulation, including the intended audience; one (1) page
      d. Hand-sketched storyboard that documents the flow and progression of the visualization with written notes; special effects, audio cues, dialogue, transitions, and scene duration should be incorporated into the storyboard; pages as needed
      e. List of references that includes sources for materials, copyrighted and otherwise; pages as needed. (The term "Fair Use" and similar terms are not acceptable citations when creating the list of references.)
      f. Permission letters for copyrighted material; pages as needed
      g. List of software and hardware used in the development of the visualization; one (1) page
      h. Student Copyright Checklist (see 2027 & 2028 Forms Appendix); one (1) page
      i. Resources and AI Reflection (see 2027 & 2028 Forms Appendix); maximum of four (4) pages

D. Recording of the Simulation:
   1. Teams will submit a URL to the video of the simulations. Entries requiring that access be granted (e.g., password protected, requires login credentials, etc.) will not be judged.
   2. The following are NOT permitted:
      a. PowerPoint presentation or PowerPoint slide show are not acceptable formats for this event.
      b. Absolutely no purchased content may be used in any part of the simulation. (Purchased content includes, but is not limited to, texture, models, and royalty free music.)
      c. Web applications that allow purchasing of elements (i.e.: Animaker and Powtoons) are not permitted.
      d. Live action video, including "whiteboard" style entries.
   3. Suggested software includes: Flash, Maya, 3DS Max, Adobe Animate, Unity, Blender, etc.
   4. Each simulation must advance automatically once it has been opened and started by the judges. A splash screen is acceptable, provided the "PLAY" command is easily visible.
   5. Simulation is not to be under two (2) or over three (3) minutes in length.
   6. A five (5) point deduction will be applied for every fifteen (15) second interval that the simulation is shorter than two (2) minutes or longer than three (3) minutes.
   7. Sound must accompany the simulation. This sound does not include a voiceover recording describing the VR simulation.

EVALUATION

PRELIMINARY ROUND

Tier 1

A. Simulation

Tier 2

B. Documentation portfolio

SEMIFINAL ROUND

A. Presentation of the simulation using participant provided Virtual Reality hardware

B. Interview about documentation and simulation.

Refer to the official rating form for more information

GENERATIVE ARTIFICIAL INTELLIGENCE

Generative Artificial Intelligence (GenAI) is permitted for this event. Refer to the Use of Generative Artificial Intelligence (GenAI) for Competitive Events found in the General Rules and Regulations.

STEM INTEGRATION

This event aligns with the STEM (Science, Technology, Engineering, and Mathematics) educational standards.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Computer animator
- Game designer
- Instructional technologist
- Software engineer

Participant/Team ID# _____

VIRTUAL REALITY SIMULATION (VR)

2027 & 2028 OFFICIAL RATING FORM

HIGH SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ TIER 1 – Simulation is playable
☐ TIER 2 – Documentation portfolio is present
☐ ENTRY NOT EVALUATED

TIER 1 – SIMULATION (90 points)				
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	Record scores in the column spaces below.
	1-4 points	5-8 points	9-10 points	
Communication of Simulation (X1)	It is difficult to understand the concept being communicated; an illogical explanation is presented.	Concept is communicated generally adequately.	Concept is communicated in an organized, clear, and concise manner.	
Creativity (X1)	Simulation lacks creativity; no, or very few, design principles are integrated in the simulation.	Some elements of creativity are expressed, with most design principles integrated.	Simulation exudes creativity; essential design principles and elements are integrated.	
Aesthetics and Artisanship (X1)	Unorganized, sloppy work is evident; simulation seems to be an afterthought and/or thrown together.	A largely organized presentation of layout and design principles is evident.	An exemplary use of layout and design principles to logically communicate important data is evident.	
Graphical Representations (X2)	Graphical representations do not help to clarify simulation, or they are of little significance to the project.	Graphical representations are appropriate and help supplement the simulation by providing clarity to the project.	Graphical representations are of excellent quality; and clarify abstract concepts.	
Video (X1)	Videos do not help to clarify simulation, or they are of little significance to the project.	Videos are appropriate and help supplement the simulation by providing clarity to the project.	Videos are of excellent quality and clarify abstract concepts.	
Sound Integration (X1)	Sounds do not help to clarify simulation, or they are of little significance to the project.	Sounds are appropriate and help supplement the simulation by providing clarity to the project.	Sounds are of excellent quality and enhance the project.	
Originality (X1)	Simulation lacks imagination, originality, and artistic detail.	Simulation is somewhat effective, inventive, and inspiring.	Simulation is inspiring, inventive, resourceful, and motivating.	
VR (X1)	Simulation lacks VR demonstration and any elements to be effecticve use of a VR simulation.	Simulation somewhat uses VR elements effectively and demonstration uses VR feature somewhat effectively	Simulation is compleatly uses VR Elements and the Demonstration is effective in showing the intergration of the hardware with the simulation	
			TIER 1 – SIMULATION SUBTOTAL (90 points)	

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	

TIME DEDUCTIONS		
There will be a five (5) point deduction for each fifteen (15) seconds under the minimum time or each fifteen (15) seconds over the maximum time allowed for the simulation.		
Total time for simulation		
Simulation time deduction		
TOTAL TIME DEDUCTION		

TIER 1 – SUBTOTAL (90 points)	

TIER 2 – DOCUMENTATION PORTFOLIO (50 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or missing three (3) or more components.	Portfolio has most components and is adequately organized.	All components are present, and content and organization are clearly evident.	
Purpose (X1)	Purpose of the simulation idea generation is unclear.	Purpose is explained appropriately and adequately.	Purpose of the simulation is clear and concisely written, and compelling.	
Storyboard (X2)	Storyboard is sloppy, seems to have been thrown together after the creation of the simulation, and/or it does not correlate with the simulation.	Storyboard is drawn appropriately and largely correlates with the completed simulation.	Storyboard is of exceptional aesthetic and artistic quality and clearly correlates with the simulation, including timings.	
Resources and AI Reflection (X1)	Form is incomplete or vague and does not clearly explain whether or how AI was used, showing little connection to the submission or understanding of effectiveness and ethics.	Form is complete and clearly explains AI use or the lack thereof with connections to the submission or theme/topic (when applicable), addressing effectiveness and ethical considerations with adequate analysis.	Form is complete, concise, and specific, critically evaluating how AI or non-AI tools shaped decisions and outcomes, with strong alignment to the submission or theme/topic (when applicable) and ethical reasoning.	
TIER 2 – DOCUMENTATION PORTFOLIO SUBTOTAL (50 points)				

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	

PRELIMINARY SUBTOTAL (140 points)	

SEMIFINAL PRESENTATION/INTERVIEW (60 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
VR Simulation (X2)	Simulation does not use the VR hardware and simulation not presented using VR formating/ features	Simulation uses the VR hardware or minimal effective use VR formating/ features.	Simulation uses the VR hardware effectively and Simulation enhanced by the use of VR formating/features.	
Organization (X1)	Team/individual seems unprepared and unorganized for the interview.	Team/individual is somewhat prepared and organized in its interview.	Team is well-prepared and any questions asked by judges are answered concisely.	
Knowledge (X2)	Team/individual seems to have little understanding of its chosen topic.	Team/individual has a generalized understanding of its chosen topic.	There is clear evidence of a thorough understanding of the chosen topic.	
Delivery (X1)	Team/individual is verbose and/or uncertain in the interview; posture, gestures, and lack of eye contact diminish the delivery.	Team/individual is somewhat well-spoken and clear in the interview; posture gestures, and eye contact result in an acceptable delivery.	Team/individual is well-spoken and distinct in the interview; posture, gestures, and eye contact result in a polished, natural, and effective delivery.	
			SEMIFINAL PRESENTATION/INTERVIEW SUBTOTAL (60 points)	

Time violation (a deduction of five (5) points total will be incurred for exceeding the semifinalist interview time limit). Record the deduction in the space to the right.	
SEMIFINAL SUBTOTAL (60 points)	
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. **TOTAL** (200 points)	

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

VIRTUAL REALITY SIMULATION (VR) EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

A. Event coordinator
B. Assistant for check-in, one (1)
C. Judges:
    1. Preliminary round, two (2) or more for initial review of entries. If more than 20 entries, provide 2 additional evaluators to conduct heats.
    2. Semifinal round, two (2) or more for interviews

MATERIALS

A. Coordinator's packet, containing:
    1. Event guidelines, one (1) copy for the coordinator and for each judge
    2. TSA Event Coordinator Report
B. Laptop
C. Stopwatch
D. Interview questions
E. Tables for entries
F. Tables and chairs for initial judges
G. Tables and chairs for semifinalist judges and participants
H. Extension cords and power-bars with protection for judges, as needed

RESPONSIBILITIES

PRE-CONFERENCE

A. National TSA will collect entries until 11:59pm ET on a designated date in mid-May. The results will be shared with the CRC manager, event coordinator, and assigned judges.
B. Review entries as they are submitted to the designated online storage utility.
C. Manage communication and Pre-conference evaluation (at least two [2] or more judges should be recruited earlier in the year). Coordinate with National TSA and/or the Judge Manager.

AT THE CONFERENCE

A. Attend the mandatory coordinator's meeting at the designated time and location.
B. Report to the CRC room and check the contents of the coordinator's packet.
C. Review the event guidelines and check to see that enough personnel have been scheduled.
D. Inspect the area or room in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
E. At least one (1) hour before the event is scheduled to begin, meet with judges/assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
F. Check that the room has the capability for participants to connect to a hotspot if needed for their VR demonstration.
G. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

A. Number of judges depends upon the number of entries. Heats of 20 entries may be used at the coordinator's discretion.
B. Judges independently assess the entries using the following procedure:
    1. Judges review and score the Simulation criteria to determine the top twenty-four (24) preliminary contestants, which will not be published.
    2. Judges score the Documentation Portfolio criteria of those top twenty-four (24) contestants to determine the top twelve (12) semifinalist teams.

C. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and a CRC manager to determine either:

   1. To deduct twenty percent (20%) of the total possible points in this round or
   2. To disqualify the entry

   The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

D. Create an interview sign-up sheet.

SEMIFINAL ROUND

A. Inspect the area in which the interviews are to take place. Ensure that there is a table and seating for the presentation/interviews.

B. At least one (1) hour before the event is scheduled to begin, meet with judges/assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.

C. Semifinalists report to the event area at the time and place stated in the conference program to sign up for a presentation/interview time.

D. Manage the presentation/interview process.

E. Discuss rule violations (e.g. 20% deduction, disqualification) and have all relevant parties initial the rating form.

F. Manage security and removal of all materials from the area.